Jacobian:

Let x, y be functions of u, v, the Jacobian of x, y w.r.t u, v is

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

The Jacobian of u, v w.r.t x, y is

$$J' = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

Let x, y, z be functions of u, v, w, the Jacobian of x, y, z w.r.t u, v, w is

$$J = \frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

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Note: (1) $JJ' = 1$

(2) If x, y are functions of u and v, u and v are functions of r and θ, then

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \frac{\partial(x,y)}{\partial(u,v)} \times \frac{\partial(u,v)}{\partial(r,\theta)}$$

(Q1) If $u = x+y+z$, $v = y+z$, $w = z$, then find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$?

$$\rightarrow \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{vmatrix}$$

$$= 1$$

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(Q2) If $x = r\cos\theta$, $y = r\sin\theta$, show that $JJ' = 1$.

$$\rightarrow J = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$= \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$J = (\cos\theta)(r\cos\theta) - (\sin\theta)(-r\sin\theta)$$

$$J = r\cos^2\theta + r\sin^2\theta$$

$$J = r(\cos^2\theta + \sin^2\theta)$$

$$J = r$$

$$r = \sqrt{x^2+y^2}$$

$$\theta = \tan^{-1}(y/x)$$

$$J' = \frac{\partial(r,\theta)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix}$$

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$$J' = \begin{vmatrix} \frac{1}{2\sqrt{x^2+y^2}}(2x+0) & \frac{1}{2\sqrt{x^2+y^2}}(+2y) \\ \frac{1}{1+(y/x)^2} \times y\left(\frac{-1}{x^2}\right) & \frac{1}{1+(y/x)^2}\left(\frac{1}{x}\right)(1) \end{vmatrix}$$

$$J' = \left[\frac{1}{2\sqrt{x^2+y^2}}(2x)\right]\left[\frac{1}{1+(y/x)^2}\right]\frac{1}{x}$$

$$- \left[\frac{1}{1+(y/x)^2} \times y \times \left(\frac{-1}{x^2}\right)\right]\left[\frac{2y}{2\sqrt{x^2+y^2}}\right]$$

$$J' = \frac{x}{\sqrt{x^2+y^2}}\left[\frac{1}{1+y^2/x^2}\right]\frac{1}{x}$$

$$+ \left[\frac{1}{1+y^2/x^2}\right]\left(\frac{y}{x^2}\right)\left(\frac{y}{\sqrt{x^2+y^2}}\right)$$

$$J' = \left[\frac{x}{\sqrt{x^2+y^2}}\right]\left[\frac{1}{\frac{x^2+y^2}{x^2}}\right]\frac{1}{x} + \left[\frac{1}{\frac{x^2+y^2}{x^2}}\right]\frac{y^2}{x^2}$$

$$\left[\frac{1}{\sqrt{x^2+y^2}}\right]$$

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$$J' = \left[\frac{1}{\sqrt{x^2+y^2}}\right]\left[\frac{x^2}{x^2+y^2}\right] + \left[\frac{y^2}{x^2+y^2}\right]\left[\frac{1}{\sqrt{x^2+y^2}}\right]$$

$$J' = \left[\frac{1}{\sqrt{x^2+y^2}}\right]\left[\frac{1}{x^2+y^2}\right]\left[x^2+y^2\right]$$

$$J' = \left[\frac{1}{\sqrt{x^2+y^2}}\right] = \frac{1}{r}$$

$$\therefore JJ' = r\left(\frac{1}{r}\right) = 1.$$

(Q2) If $u = x^2 - 2y^2$, $v = 2x^2 - y^2$, $x = r\cos\theta$ and $y = r\sin\theta$, then show that

$$\frac{\partial(u,v)}{\partial(r,\theta)} = 6r^3 \sin 2\theta$$

$$\rightarrow \quad \frac{\partial(u,v)}{\partial(r,\theta)} = \frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(r,\theta)}$$

$$\frac{\partial(u,v)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

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$$\frac{\partial(u,v)}{\partial(r,\theta)} = \begin{vmatrix} 2x & -4y \\ 4x & -2y \end{vmatrix} \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$= [2x(-2y) - 4x(-4y)][\cos\theta(r\cos\theta) - \sin\theta(-r\sin\theta)]$$

$$= [-4xy + 16xy][r\cos^2\theta + r\sin^2\theta]$$

$$= 12xy\,r(\cos^2\theta + \sin^2\theta)$$

$$= 6 \times 2 \times r\cos\theta \times r\sin\theta \times r \times 1$$

$$= 6r^3 \sin 2\theta$$

(Q3) If $u = x+y+z$, $v = y+z$ and $w = z$, then find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$?

→ $$J = \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$J = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

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(Q4) If $u = \frac{yz}{x}$, $v = \frac{xz}{y}$, $w = \frac{xy}{z}$, then find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$?

$$\rightarrow \quad J = \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$J = \begin{vmatrix} -\frac{yz}{x^2} & \frac{z}{x} & \frac{y}{x} \\ \frac{z}{y} & -\frac{xz}{y^2} & \frac{x}{y} \\ \frac{y}{z} & \frac{x}{z} & -\frac{xy}{z^2} \end{vmatrix}$$

$$J = \left(\frac{-yz}{x^2}\right)\left[\left(\frac{-xz}{y^2}\right)\left(\frac{-xy}{z^2}\right) - \left(\frac{x}{z}\right)\left(\frac{x}{y}\right)\right]$$

$$- \frac{z}{x}\left[\left(\frac{z}{y}\right)\left(\frac{-xy}{z^2}\right) - \left(\frac{y}{z}\right)\left(\frac{x}{y}\right)\right]$$

$$+ \frac{y}{x}\left[\left(\frac{z}{y}\right)\left(\frac{x}{z}\right) - \left(\frac{y}{z}\right)\left(\frac{-xz}{y^2}\right)\right]$$

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$$J = -\frac{yz}{x^2}\left[\frac{x^2}{yz} - \frac{x^2}{yz}\right] - \frac{z}{x}\left[-\frac{x}{z} - \frac{x}{z}\right]$$

$$+ \frac{y}{x}\left[\frac{x}{y} + \frac{x}{y}\right]$$

$$J = -\frac{yz}{x^2}(0) - \frac{z}{x}\left[-\frac{2x}{z}\right] + \frac{y}{x}\left[\frac{2x}{y}\right]$$

$$J = 0 + 2 + 2 = 4$$

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Q5 If $u = x + \frac{y^2}{x}$, $v = \frac{y^2}{x}$, then show that $JJ' = 1$.

$$\rightarrow J = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$J = \begin{vmatrix} 1 + y^2\left(\frac{-1}{x^2}\right) & 0 + \frac{1}{x}(2y) \\ y^2\left(\frac{-1}{x^2}\right) & \frac{1}{x}(2y) \end{vmatrix}$$

$$J = \left[1 - \frac{y^2}{x^2}\right]\left[\frac{2y}{x}\right] - \left[\frac{-y^2}{x^2}\right]\left[\frac{2y}{x}\right]$$

$$J = \left[\frac{2y}{x}\right]\left[1 - \frac{y^2}{x^2} + \frac{y^2}{x^2}\right]$$

$$J = \frac{2y}{x}$$

$$u = x + \frac{y^2}{x}, \quad v = \frac{y^2}{x}$$

$$\Rightarrow u = x + v$$

$$\Rightarrow \boxed{x = u - v}$$

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$$v = \frac{y^2}{x} \Rightarrow y^2 = vx \Rightarrow y^2 = v(u-v)$$

$$y = \sqrt{v(u-v)} = \sqrt{uv - v^2}$$

$$J' = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$J' = \begin{vmatrix} 1 & -1 \\ \frac{1}{2\sqrt{uv-v^2}}(v) & \frac{1}{2\sqrt{uv-v^2}}(u-2v) \end{vmatrix}$$

$$J' = \frac{1}{2\sqrt{uv-v^2}}[u-2v] - \left[\frac{-v}{2\sqrt{uv-v^2}}\right]$$

$$J' = \frac{1}{2\sqrt{uv-v^2}}[u-2v+v]$$

$$J' = \frac{u-v}{2\sqrt{uv-v^2}} = \frac{x}{2y}$$

$$\therefore JJ' = \left(\frac{2y}{x}\right)\left(\frac{x}{2y}\right) = 1$$

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